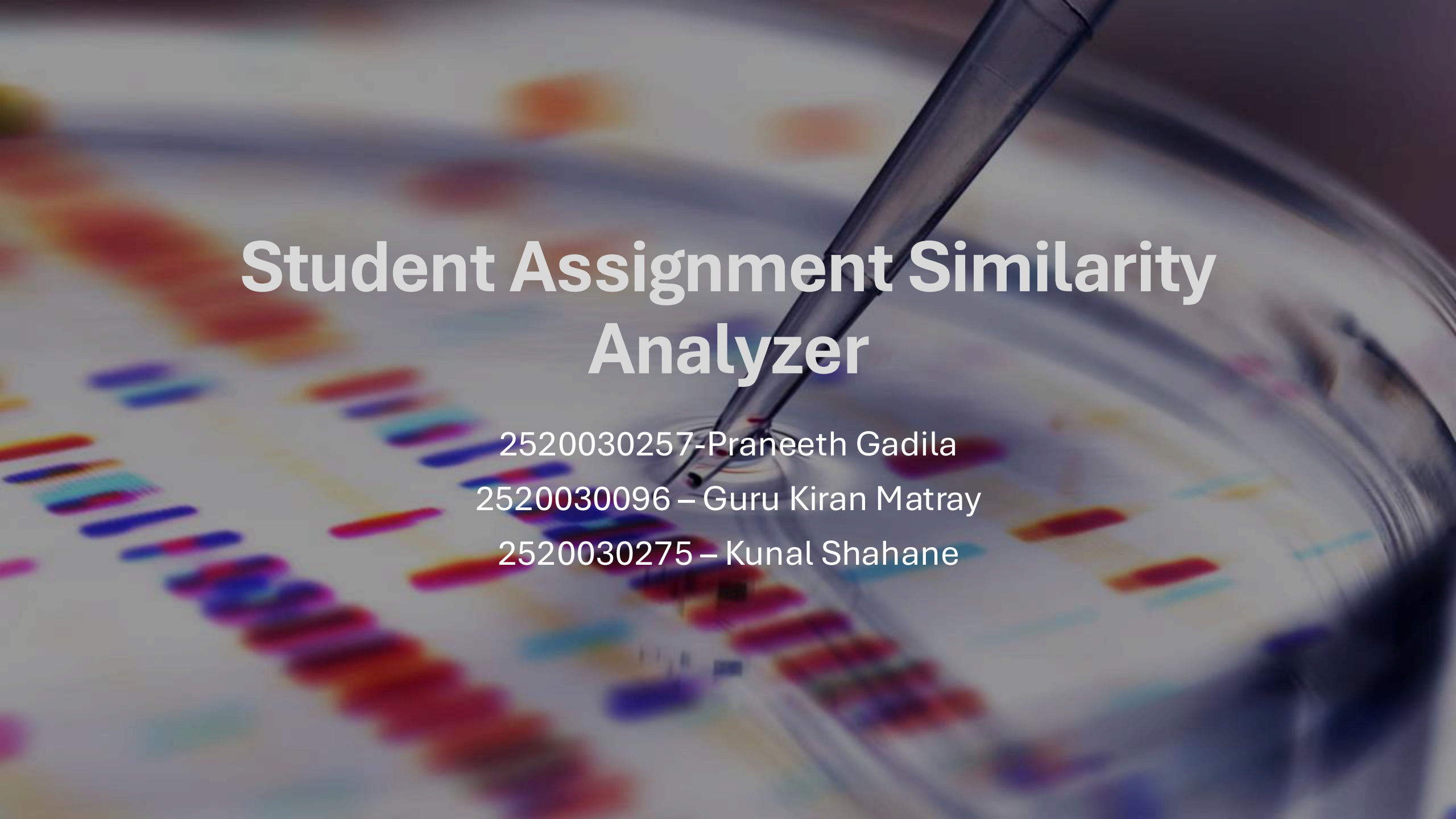


The background of the slide is a blurred image of a DNA microarray. It shows a grid of small, colorful spots (red, green, blue, yellow) on a white surface. A glass pipette tip is visible in the upper right, pointing towards the center of the array. The overall image has a soft, out-of-focus quality.

Student Assignment Similarity Analyzer

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Abstract & Problem Statement

ABSTRACT

- A Java-based system for detecting plagiarism and similarity in **student TXT/PDF assignments**.
- Automatically compares each new submission with **all previous assignments**.
- Uses **String Algorithms and Dynamic Programming** to detect exact and modified copied content.
- Generates similarity percentages and identifies the most similar submissions.

PROBLEM STATEMENT

- Manual comparison of multiple assignments is **time-consuming and difficult to scale**.
- Students may modify copied content through **word changes, insertions, deletions, or transpositions**.
- An automated system is needed to efficiently compare submissions and flag suspicious similarities.
- The system classifies assignments using a **10% similarity threshold: $\leq 10\%$ Original, $> 10\%$ Copied**.

Description

- Java-based **console application** for detecting assignment similarity and plagiarism.
- Accepts **TXT and PDF** assignment files.
- Extracts and preprocesses text from submitted documents.
- Uses **TF-IDF and Cosine Similarity** to calculate document similarity.
- Compares new assignments with previously submitted reference documents.
- Generates a **similarity percentage** for each comparison.
- Classifies assignments as **Original or Copied** using a predefined threshold.
- Demonstrates practical application of **Data Structures & Algorithms**.
- **Technologies:** Java 17+, Apache PDFBox, TF-IDF, Cosine Similarity.

Tools, Technologies & Expected Outcome

Tools & Technologies

- **Language:** Java
- **GUI:** Java Swing
- **IDE:** Visual Studio Code
- **File Formats:** TXT & PDF
- **PDF Processing:** Apache PDFBox
- **String Algorithms:** Naïve, KMP, Z-Function, Rabin-Karp, Aho-Corasick
- **Suffix Algorithms:** Suffix Array, LCP & Kasai
- **Dynamic Programming:** Levenshtein, Wagner-Fischer, Damerau-Levenshtein
- **Similarity:** TF-IDF & Cosine Similarity
- **Data Structures:** Trie, HashMap, Arrays, Lists

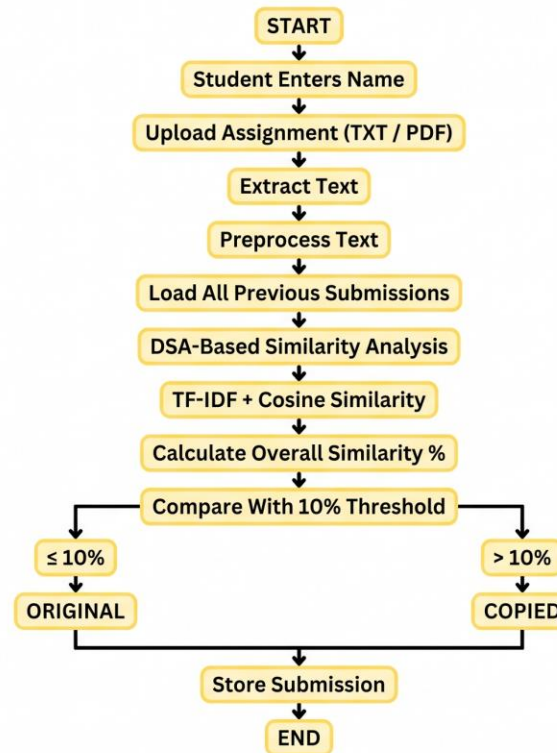
Tools, Technologies & Expected Outcome

Expected Outcome

- Automatic comparison with **all previous assignments**.
- Accurate identification of **exact and modified copied content**.
- Pairwise similarity percentage for every previous submission.
- Identification of the **highest-similarity assignment**.
- Automatic **Original/Copied classification** using the 10% threshold.
- Reduced manual plagiarism checking and improved scalability for large numbers of submissions.

Workflow

STUDENT ASSIGNMENT SIMILARITY ANALYZER



Geotag images

